

NeuroCurriculum: Translating Neurodiversity Theory into Adaptive Curriculum Design

School of Business · University at Albany, State University of New York
Assistive Technology Innovation Center (ATIC)



UNIVERSITY
AT ALBANY
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1 The Problem

One-size-fits-all materials leave many learners behind.



ADHD Autism Dyslexia

Materials don't match how learners process information

Neurodivergent learners, including those with attention deficit hyperactivity disorder (ADHD), autism, and dyslexia, may face challenges when traditional instructional materials do not align with how they process and engage with information.

Personalization takes time

However, creating personalized learning materials can require significant time and effort from educators.

2 Our Solution

An AI-supported curriculum adaptation framework

As part of the Assistive Technology Innovation Center (ATIC) initiative, we are developing an artificial intelligence (AI)-supported curriculum adaptation framework that preserves learning objectives while adapting the structure, explanation, and presentation of instructional content.

WHAT CHANGES · WHAT STAYS

Structure Explanation Presentation

✓ Learning objectives preserved

OUR GOAL

Reduce the time required to create accessible and personalized learning materials based on different cognitive and learning needs.

THE STUDY ASKS HOW TO

Translate neurodiversity theory into reusable adaptation mechanisms.

We are working with educators, disability-service professionals, caregivers, and accessibility experts to ensure that the framework is practical, evidence-informed, and relevant to real educational and training settings.

3 Four Design Principles

- 01 Preserve instructional intent**
Keep the source meaning, required concepts, and learning goals while changing delivery.
- 02 Adapt to functional needs**
Use ECC, LIP, and SER characteristics instead of diagnosis alone.
- 03 Keep adaptation model-neutral**
Specify rules and output schema outside provider-specific implementations.
- 04 Put educators in control**
Review and revise the editable curriculum before learners see rendered materials.

4 Personalization for Neurodiverse Learners

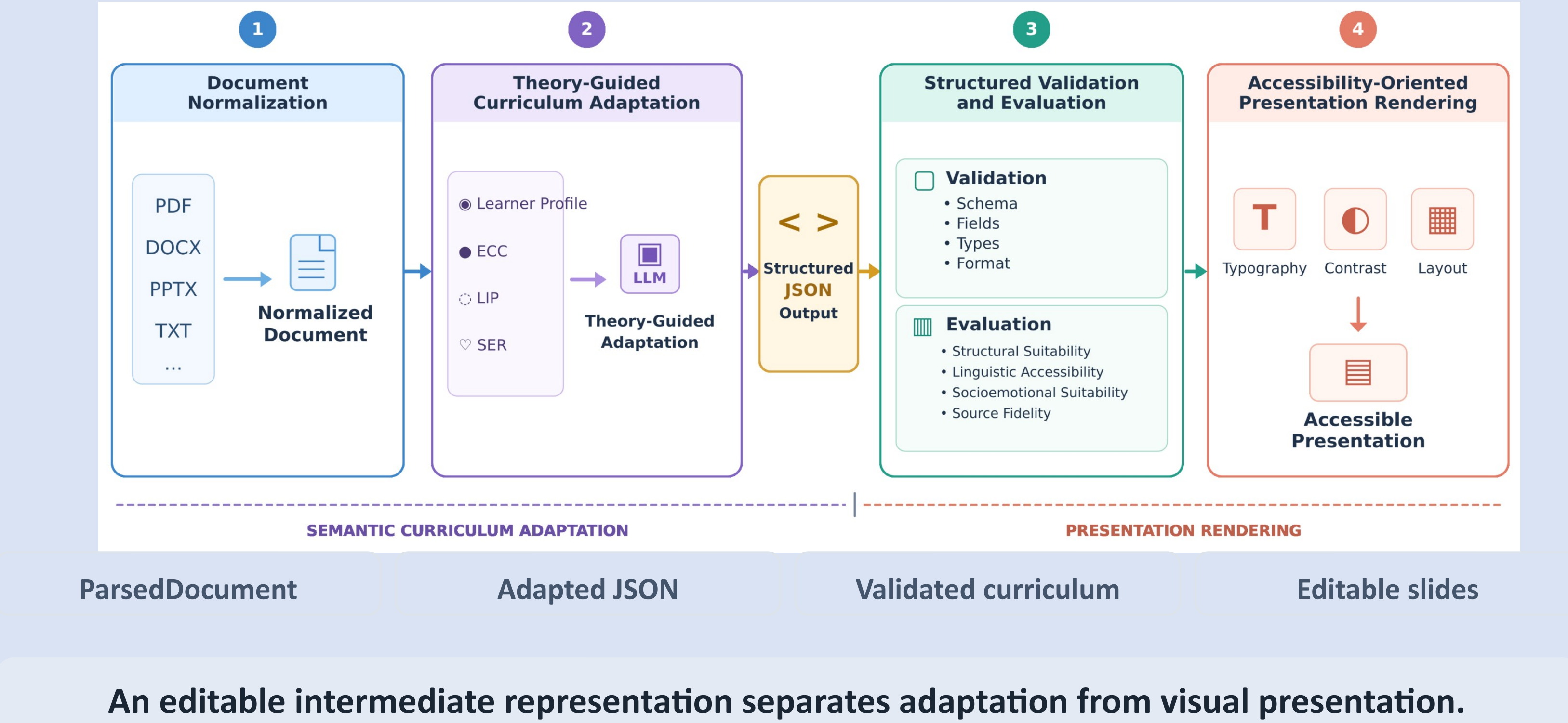
(1) NEURODIVERGENT TYPE	(2) AREA OF DIFFICULTY	(3) CURRICULUM VARIABLE
ADHD Attention / executive function	Attention	Focused tasks
	Working memory	Visual reminders
	Time management	Visual timers
ASD Autism spectrum	Literalism	Plain language
	Sensory overload	Clear structure
Dyslexia Reading access	Reading fluency	Chunked text
	Visual stress	Readable spacing

Illustrative mappings summary; choose supports for each learner with educator review.

5 From Learning Needs to Adaptation

CROSS-CONDITION DIMENSION	UDL ALIGNMENT	CURRICULUM MECHANISMS
ECC Executive & Cognitive Control	Action & Expression	Chunk content; break tasks into steps; add checkpoints and pacing cues.
LIP Linguistic & Information Processing	Representation	Define terms; shorten sentences; segment explanations and examples.
SER Socio-Emotional & Sensory Regulation	Engagement	Use supportive language; reduce pressure and sensory load.

6 NeuroCurriculum Workflow



7 Computational Demonstration & Results

3 Source documents
DOCX PDF PPT

3 Learner profiles
ADHD · ASD · dyslexia

4 Language models
ChatGPT Claude LLaMA Gemini

36 conditions · One generation per condition
· Same prompt and schema

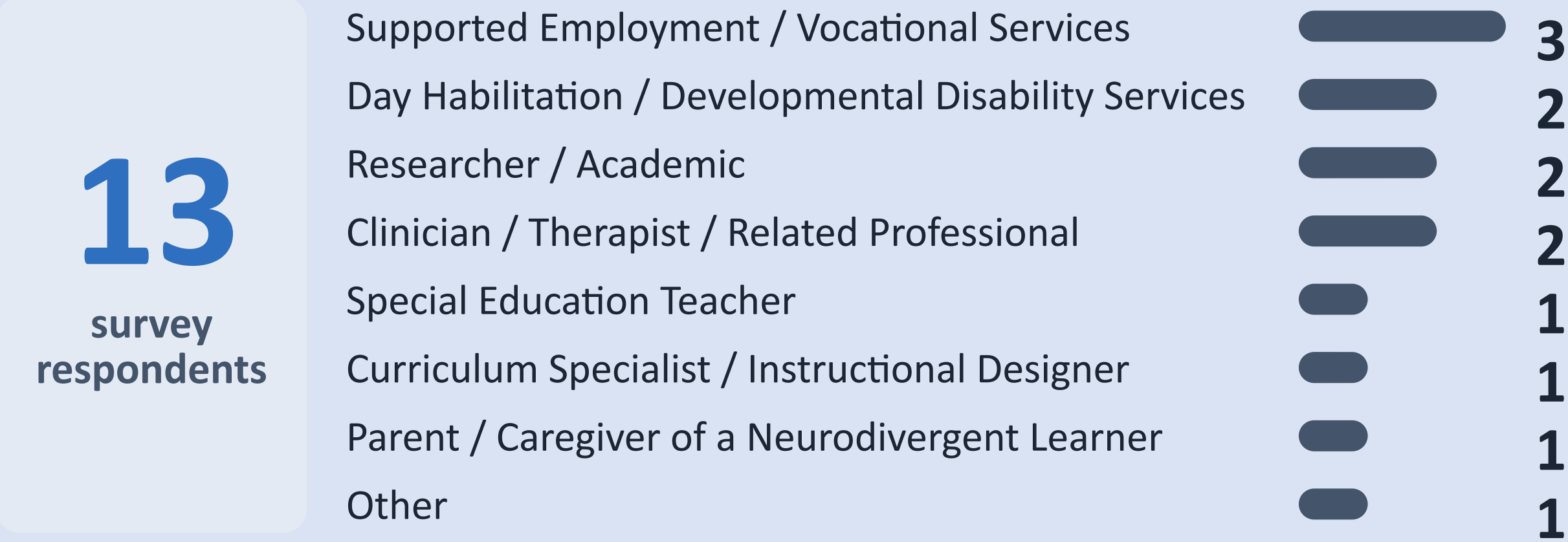
Mean scores by model (0–1)

Model	ECC	LIP	SER	Fidelity	Overall
GPT-4o	.697	.614	.764	.655	.674
Claude Sonnet 5	.430	.524	.792	.882	.643
Gemini 2.5 Pro	.491	.560	.792	.738	.627
Llama 3.1 8B	.506	.485	.722	.910	.654



Source fidelity: Mean scores range from .655 to .910 across providers while structure changes.
Model differences: Providers show distinct strengths across ECC, LIP, SER, and fidelity.
Design science contribution: A reusable artifact connects neurodiversity theory, UDL, curriculum transformation, and evaluation in one inspectable workflow.

8 Expert & Stakeholder Evaluation



OUTREACH GROUPS

Neurodivergent Individuals · Direct lived-experience participation to be expanded
Caregivers / Family · NYS Executive Chamber / Andy Sink
Special Education & Academic Professionals · University at Albany; Down Syndrome Aim High Resource Center; Measurement Incorporated
Disability-Service Practitioners · Community Services for Every1; Liberty; Catholic Charities; Arc organizations; Heartshare; Living Resources; Abilities First; Pathfinder Village

NEXT OUTREACH PRIORITY

Increase direct participation from neurodivergent individuals and people with developmental disabilities.

9 Video & Real-Time Demo



Watch the demo video



Try it yourself!